YUCHEOL SHIN

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EDUCATION

Ongoing **Ph.D. Comparative Biology** **Advisor: Dr. Frank T. Burbrink**
Richard Gilder Graduate School, American Museum of Natural History (AMNH), New York, NY, USA

2016 – 2024 **B.S. Biological Sciences** **Advisor: Dr. Daesik Park**
Kangwon National University, Chuncheon, South Korea
Thesis: Pleistocene range dynamics of the two distantly related lungless salamanders endemic to the Korean Peninsula

2018 **Study Abroad, Biology** **Advisor: Dr. Jenny Duggan**
California State University, Monterey Bay, Seaside, CA, USA
Project: Comparing the efficiency of different survey methods for detecting herpetofauna at the UC Fort Ord Natural Reserve

PUBLICATIONS

(All publications are available from: https://www.researchgate.net/profile/Yucheol_Shin)

** denotes co-first author papers*

Research articles

19. A Borzée, A Angulo, H Meredith, J Groffen, ….., **Y Shin,** Y Bae, Z Wang, Z Qiu, R Pearce. *In review*. Protecting Japanese Giant Salamanders (*Andrias japonicus*) in the Nawa River Basin, Japan: policy recommendations addressing water pollution and waterway disruption. Frontiers in Amphibian and Reptile Science.

18. JY Jeon, **Y Shin**, AJ Mularo, X Feng, JA DeWoody. 2024. The integration of whole-genome resequencing and ecological niche modeling to conserve profiles of local adaptation. Diversity and Distributions 30: e13847.

17. I-K Park, **Y Shin,** H-J Baek, J Kim, D-I Kim, M Seok, Y Oh, D Park. 2024. Establishment potential across South Korea for two gecko species, *Gekko japonicus* and *G. swinhonis*, adapted to different climates. NeoBiota 93: 39-62.

16. A Borzée, **Y Shin**, Y Bae, D Jeong, H Amin, M-S Min, SN Othman. 2024. From Korean to northeast Asian endemicity: on the occurrence of *Pelophylax chosenicus* along the Eastern Coastal Yellow Sea. Frontiers of Biogeography 16: e62301.

15. **Y Shin***, K Heo*, SN Othman, Y Jang, M-S Min, A Borzée. 2024. Tracing the geographic origin of a nonnative Red-banded Snake (Colubridae: *Lycodon rufozonatus*) found on Jeju Island, Republic of Korea. Herpetologica 80: 30-39.

14. Y Peng, Y Li, G Cao, H Li, **Y Shin**, Z Piao, F Perez, W Zhu, A Borzée. 2023. Estimation of habitat suitability and landscape connectivity for Liaoning and Jilin clawed salamanders (Hynobiidae: *Onychodactylus*) in the transboundary region between the People's Republic of China and the Democratic People's Republic of Korea. Global Ecology and Conservation 48: e02694.

13. J Li, **Y Shin**, M Hong, Y Jang, A Borzée. 2023. Is ultra-violet fluorescence a trait related to breeding in the Mongolian racerunner (*Eremias argus*; Lacertidae, Reptilia)? Journal of Biological Research – Thessaloniki 30:3.

12. A Borzée, M Rodriguez, N Bhatri, YI Kim, ….., VK Prasad, Y Jang, **Y Shin**, SN Othman. 2023. Policy recommendation on the *Rana* trade towards the Republic of Korea. Frontiers in Environmental Science 11: 1097849.

11. D Jeong, **Y Shin**, B Lim, H Serret, Y Jang. 2023. Do Barn Swallows (*Hirundo rustica gutturalis*) prefer to breed in inhabited houses in South Korea? The Wilson Journal of Ornithology 134: 633-641.

10. **Y Shin**, K Kim, J Groffen, D Woo, E Song, A Borzée. 2022. Citizen science and roadkill trends in the Korean herpetofauna: the importance of spatially biased and unstandardized data. Frontiers in Ecology and Evolution 10: 944318.

9. SN Othman, **Y Shin**, H-T Kim, M-F Chuang, Y Bae, J Hoti, Y Zhang, Y Jang, A Borzée. 2022. Evaluating the efficiency of popular species identification methods, and comparative workflow using morphometry and barcoding bioinformatics for taxonomy and origin of traded cryptic brown frogs. Global Ecology and Conservation 38: e02253.

8. A Borzée, **Y Shin**, N Poyarkov, JY Jeon, HJ Baek, CH Lee, J An, YJ Hong, M-S Min. 2022. Dwindling in the mountains: description of a critically endangered and microendemic *Onychodactylus* species (Amphibia, Hynobiidae) from the Korean Peninsula. Zoological Research 43: 750−755.

7. **Y Shin**, M-S Min, A Borzée. 2021. Driven to the edge: Species distribution modeling of a Clawed Salamander (Hynobiidae: *Onychodactylus koreanus*) predicts range shifts and drastic decrease of suitable habitats in response to climate change. Ecology and Evolution 11: 14669-14688.

6. A Borzée, SN Litvinchuk, K Ri, D Andersen, TY Nam, GH Jon, HS Man, JS Choe, S Kwon, SN Othman, K Messenger, Y Bae, **Y Shin**, A Kim, I Maslova, J Luedtke, L Hobin, N Moores, B Seliger, F Glenk, Y Jang. 2021. Update on distribution and threat status of amphibians in the Democratic People's Republic of Korea: conclusions based on field surveys, landscape modeling, molecular analyses, and call properties. Animals 11:2057.

5. **Y Shin**, Y Jang, A Borzée. 2021. Snakebite envenomings in the Republic of Korea from the 1970s to the 2020s: a review. Toxicon 196: 8-18.

4. **Y Shin***, KR Messenger*, KS Koo, SC Lee, M Hou, A Borzée. 2021. How threatened is *Scincella huanrenensis*? An update on threats and trends. Conservation 1: 58-72.

3. D Macias, **Y Shin**, A Borzée. 2021. An update on the conservation status and ecology of Korean terrestrial squamates. Journal for Nature Conservation 60: 125971.

2. **Y Shin**, Y Jang, SJR Allain, A Borzée. 2020. Catalogue of herpetological specimens of the Ewha Womans University Natural History Museum (EWNHM), Republic of Korea. ZooKeys 965: 103-139.

1. A Borzée, KR Messenger, S Chae, D Andersen, J Groffen, YI Kim, J An, SN Othman, K Ri, TY Nam, Y Bae, J-L Ren, J-T Li, M-F Chuang, Y Yi, **Y Shin**, T Kwon, Y Jang, M-S, Min. 2020. Yellow sea mediated segregation between North East Asian *Dryophytes* species. PLoS ONE 15: e0234299.

Book chapters

3. A Borzée, K Milto, L Borkin, X Ji, W Zhao, **Y Shin.** 2022. *Takydromus wolteri*. In: The IUCN Red List of Threatened Species 2022 : e.T47756012A214715587.

2. K Milto, L Borkin, X Ji, W Zhao, **Y Shin**. 2021. *Takydromus wolteri*. In: The IUCN Red List of Threatened Species 2021: e.T47756012A47756021.

1. V Orlova, K Milto, L Borkin, W Zhao, **Y Shin**. 2021. *Takydromus amurensis*. In: The IUCN Red List of Threatened Species 2021: e.T47755997A47756006.

Short communications

7. I-K Park, **Y Shin,** D Park. *In press*. Predation on a metamorphosing tadpole of the American Bullfrog, *Aquarana catesbeiana* (Shaw, 1802), by the fishing spider *Dolomedes sulfureus*, in the Republic of Korea. Herpetology Notes.

6. **Y Shin**, J Miller, JM Duggan. 2021. Observations on the year-round communal use of an artificial structure by Northern Pacific rattlesnakes (*Crotalus oreganus oreganus*) in coastal Central California. Herpetological Bulletin 155: 25-27.

5. **Y Shin**, A Borzée. 2020. Melanism in the Ussuri pitviper (*Gloydius ussuriensis*) from the Republic of Korea, with remarks on color variations. Jordan Journal of Natural History 7: 60-63.

4. **Y Shin**, J Ambu, A Borzée. 2020. Observations on heterospecific amplexus in Asiatic toads (Anura: Bufonidae: *Bufo gargarizans*) in the Republic of Korea. Herpetology Notes 13: 411-413.

3. **Y Shin**, D Jeong, A Borzée. 2020. Mass displacement of Korean clawed salamanders (*Onychodactylus koreanus*) and the threat of road-kill. Herpetological Bulletin 151: 28-31.

2. **Y Shin**, Y Jang, T Kim, A Borzée. 2020. A specimen of *Karsenia koreana* (Caudata: Plethodontidae) misidentified as *Hynobius leechii* 27 years before the species' description and additional historical record. Current Herpetology 39: 75-79.

1. **Y Shin**, Y Jang, A Borzée. 2020. Limb malformations in *Bombina orientalis* (Anura; Bombinatoridae) in the Republic of Korea based on museum specimens. Herpetology Notes 13: 29-31.

Natural history & distribution notes

8. J-J Strydom, **Y Shin**. 2022. *Gloydius brevicauda* (Short-tailed Pitviper). Diet. Herpetological Review 53: 697-698.

7. K Heo, **Y Shin**, K Messenger. 2022. *Hynobius notialis* (Southern Korean Salamander). Behavior. Herpetological Review 53: 642.

6. D Jeong, B Lim, **Y Shin**. 2022. *Dryophytes japonicus* (Japanese Treefrog). Predation. Herpetological Review 53: 467-468.

5. **Y Shin**, D Jeong, B Lim. 2021. *Kaloula borealis* (Boreal Digging Frog). Hibernaculum. Herpetological Review 52: 828-829.

4. **Y Shin**, MD Cho, A Borzée. 2021. *Gloydius ussuriensis* (Ussuri Pitviper). Diet. Herpetological Review 52: 157-158.

3. K Messenger, H Dahn, D Andersen, **Y Shin**, Y Yi, Y Bae, A Borzée. 2020. *Bufo stejnegeri* (Korean Water Toad). Geographical distribution. Herpetological Review 51: 532.

2. **Y Shin**, Y Jang, A Borzée. 2020. *Gloydius saxatilis* (Rock Mamushi). Dicephalism. Herpetological Review 51: 143-144.

1. **Y Shin**, Y Jang, A Borzée. 2020. *Gloydius ussuriensis* (Ussuri Mamushi). Disease. Herpetological Review 51: 144.

PRESENTATIONS

** denotes the presenter. All delivered in English unless stated otherwise.*

2024 **Y Shin***, A Borzée. Establishment potential of non-native brown frogs (*Rana*) in the Republic of Korea. Poster. 15th Student Conference on Conservation Science-New York (SCCS-NY). American Museum of Natural History, New York, USA.

2024 **Y Shin***, A Borzée, D Park. Climatic data sources influence the prediction of past habitat suitability and niche comparisons for two distantly related lungless salamanders. Oral presentation. 10th World Congress of Herpetology. Kuching, Sarawak, Borneo.

2024 **Y Shin***, SN Othman, DB Kohler, I Maslova, K Messenger, A Borzée. Contributions to the knowledge of pitvipers (genus *Gloydius*) in the Democratic People's Republic of Korea. Poster. 10th World Congress of Herpetology. Kuching, Sarawak, Borneo.

2024 H Amin*, SN. Othman, **Y Shin**, M-S Min, KR Messenger, A Movahedi, D Ghosh, DB Kohler, VK Prasad, X Zhang, Y Yi, Y Jang, H Ruan, A Borzée. An integrative analysis for the reconsideration of species boundaries in East Asian *Pelophylax* species. Oral presentation. 10th World Congress of Herpetology. Kuching, Sarawak, Borneo.

2023 **Y Shin***. Where did you come from? Using molecular phylogenetics to trace the geographic origin of a non-native Red-banded Snake (*Lycodon rufozonatus*) found on Jeju Island, Republic of Korea. Oral presentation [in Korean]. KNU College of Natural Sciences Undergraduate Research Competition.

2023 **Y Shin***. Pleistocene range dynamics of the two distantly related lungless salamanders endemic to the Korean Peninsula. Oral presentation [in Korean]. KNU Department of Biological Sciences Senior Research Thesis Presentation.

2023 **Y Shin***, D Park. Pleistocene range dynamics of the two distantly related lungless salamanders endemic to the Korean Peninsula. Poster. The 10th EAFES International Congress. Jeju, Republic of Korea.

2022 **Y Shin***, K Kim, J Groffen, A Borzée. Citizen science and roadkill trends in the Korean herpetofauna. Poster. 13th Student Conference on Conservation Science-New York (SCCS-NY). American Museum of Natural History, New York, USA.

2021 **Y Shin***, M-S Min, A Borzée. Distribution modeling of *Onychodactylus koreanus* predicts drastic decrease of suitable habitats in response to climate change. Poster. Joint Meeting of Ichthyologists and Herpetologists. Phoenix, Arizona, USA **(SSAR George B. Rabb Undergraduate Poster Award)**.

2021 JY Jeon*, **Y Shin**, M-S Min, DK Lee. The effects of climate change on the distribution of Gori Salamander (*Hynobius yangi*). Oral presentation [in Korean]. Meetings of the Korean Society of Climate Change Research. Suwon, Republic of Korea **(Best Poster Presentation Award)**.

2020 **Y Shin***, Y Jang, A Borzée. A metanalysis of snakebites in South Korea: bridging the knowledge gap in snakebite envenomation. Poster. 9th World Congress of Herpetology. Dunedin, New Zealand.

2018 **Y Shin***, JM Duggan. Comparing survey methods and habitat associations for herpetofauna at the UC Fort Ord Natural Reserve. Poster. The Wildlife Society Annual Wildlife Symposium. San Luis Obispo, California, USA.

2018 **Y Shin***, JM Duggan. Comparing efficiency of different survey methods for detecting snakes at the UC Fort Ord Natural Reserve: preliminary observations. Poster. 5th Annual Summer Research Symposium, California State University, Monterey Bay. Seaside, California, USA.

RESEARCH EXPERIENCES

2022 – 2024 **Undergraduate Researcher** **Advisor: Dr. Daesik Park**

Laboratory of Herpetology, Kangwon National University

- Investigation of contemporary niche overlaps, Pleistocene range dynamics, and range formation processes of Korean endemic salamanders *Onychodactylus koreanus* and *Karsenia koreana*.
- Investigation of courtship behaviors in the Asian plethodontid *Karsenia koreana*.
- Participated in herpetological surveys across South Korea.

2019 – 2024 **Research Assistant** **Advisor: Dr. Amaël Borzée**

Laboratory of Animal Behaviour and Conservation, Nanjing Forestry University

- Ecological niche modeling of northeast Asian herpetofauna.
- Collaboration in multiple projects on the systematics, ecology, and conservation of northeast Asian herpetofauna.

2019 **Herpetology Curatorial Assistant** **Advisors: Dr. Amaël Borzée, Dr. Yikweon Jang**

Ewha Womans University Natural History Museum (EWNHM)

- In charge of photographing and cataloging 1554 formalin-fixed specimens deposited in the herpetological collection.

2019 **Research Intern** **Advisors: Dr. Amaël Borzée, Dr. Yikweon Jang**

Laboratory of Animal Communication, Ewha Womans University

- Collaboration in multiple projects on the systematics, ecology, and conservation of northeast Asian herpetofauna.
- Participated in herpetological surveys across South Korea.

2018 **Undergraduate Researcher** **Advisor: Dr. Jenny Duggan**

Terrestrial Wildlife Ecology Lab, California State University, Monterey Bay

- Conducted herpetofauna surveys using coverboards and drift fence arrays on the UC Fort Ord Natural Reserve during 193 survey days.

FELLOWSHIPS, GRANTS, AND AWARDS

2023 Excellent Presentation Award (200,000 KRW). KNU College of Natural Sciences Undergraduate Research Competition

2022 SSAR Roger Conant Grant-in-Herpetology, Undergraduate Research (500 USD). Society for the Study of Reptiles and Amphibians

2022 KNU Foundation Scholarship (1,140,500 KRW). Kangwon National University

2022 Scholarship for Outstanding Academic Performance (862,000 KRW). Kangwon National University

2021 George B. Rabb Undergraduate Poster Award (250 USD). Society for the Study of Amphibians and Reptiles

2021 Best Poster Presentation Award. The Korean Society of Climate Change Research

2018 Undergraduate Summer Research Equipment Grant (390 USD). Undergraduate Research Opportunities Center, California State University, Monterey Bay

2018 KNU Learning Core Program Student Mentor Award (650,000 KRW). Kangwon National University

2018 BEST KNU Scholarship - Highest academic honors (2,101,000 KRW). Kangwon National University

2017 Scholarship for Outstanding Academic Performance (1,725,000 KRW). Kangwon National University

2017 BEST KNU Scholarship - Highest academic honors (2,101,000 KRW). Kangwon National University

2016 BEST KNU Scholarship - Highest academic honors (2,101,000 KRW). Kangwon National University

2016 Freshmen English Proficiency Award - 1st place (100,000 KRW). Kangwon National University

RESEARCH SKILLS

- Ecological niche modeling and niche analyses in R
- Data processing, statistical analyses, and data visualization in R
- Geospatial analyses and visualization in QGIS and ArcMap
- Phylogenetic analyses